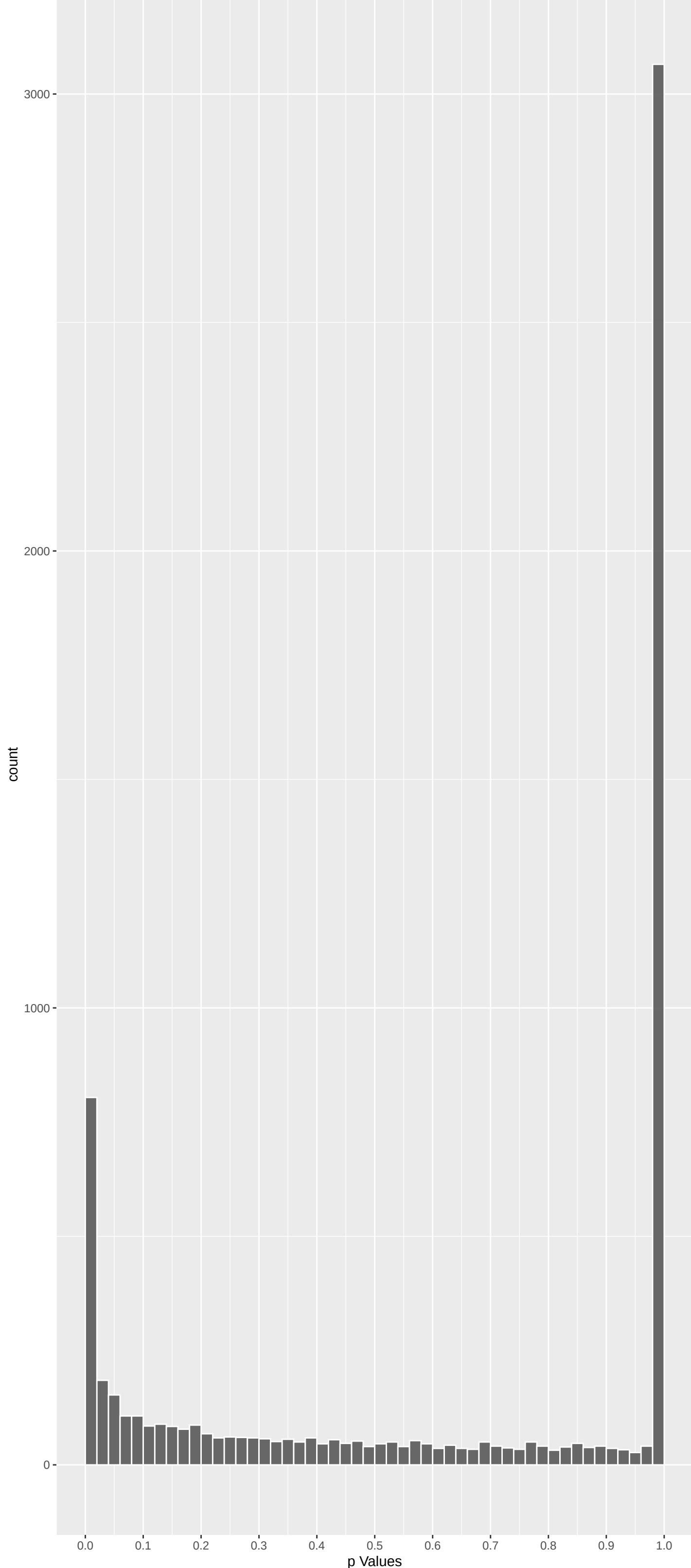


Top genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
SPG11	-7.877562	1.001490e-14	1.616e-10	2.606e-06
PLEC	7.335147	6.644363e-13	3.573e-09	1.921e-05
TMEM131L	-7.338137	6.497642e-13	3.573e-09	1.921e-05
ADGRF1	-7.170844	2.236109e-12	9.018e-09	3.291e-05
SLC14A2	-7.103732	3.642979e-12	1.175e-08	3.291e-05
CLCNKB	-6.959520	1.024299e-11	1.836e-08	3.291e-05
MMEL1	6.980375	8.831825e-12	1.836e-08	3.291e-05
EP400	6.964074	9.917134e-12	1.836e-08	3.291e-05
TRPM5	7.016268	6.836189e-12	1.836e-08	3.291e-05
AGGF1	-6.891699	1.653897e-11	2.668e-08	4.304e-05
PJA2	-6.856022	2.124129e-11	3.115e-08	4.568e-05
ECI2	-6.761969	4.083872e-11	5.490e-08	7.381e-05
TACR2	6.695199	6.461332e-11	8.018e-08	9.950e-05
CLCNKA	-6.668773	7.738535e-11	8.917e-08	1.027e-04
ADGRV1	-6.625078	1.041198e-10	1.120e-07	1.204e-04
ZFAND4	-6.571942	1.489891e-10	1.358e-07	1.217e-04
CFAP20DC	-6.585897	1.356439e-10	1.358e-07	1.217e-04
SFSWAP	6.569396	1.515589e-10	1.358e-07	1.217e-04
MTBP	-6.544606	1.789571e-10	1.519e-07	1.236e-04
TIGD4	-6.535614	1.900473e-10	1.533e-07	1.236e-04
ZNF292	-6.481471	2.724982e-10	2.080e-07	1.525e-04
VEPH1	-6.475435	2.836180e-10	2.080e-07	1.525e-04
RAVER2	-6.310294	8.355200e-10	5.860e-07	4.110e-04
PATL2	-6.287553	9.675246e-10	6.503e-07	4.134e-04
AKAP9	-6.278581	1.025034e-09	6.614e-07	4.134e-04

Positive Rho Non-permuted



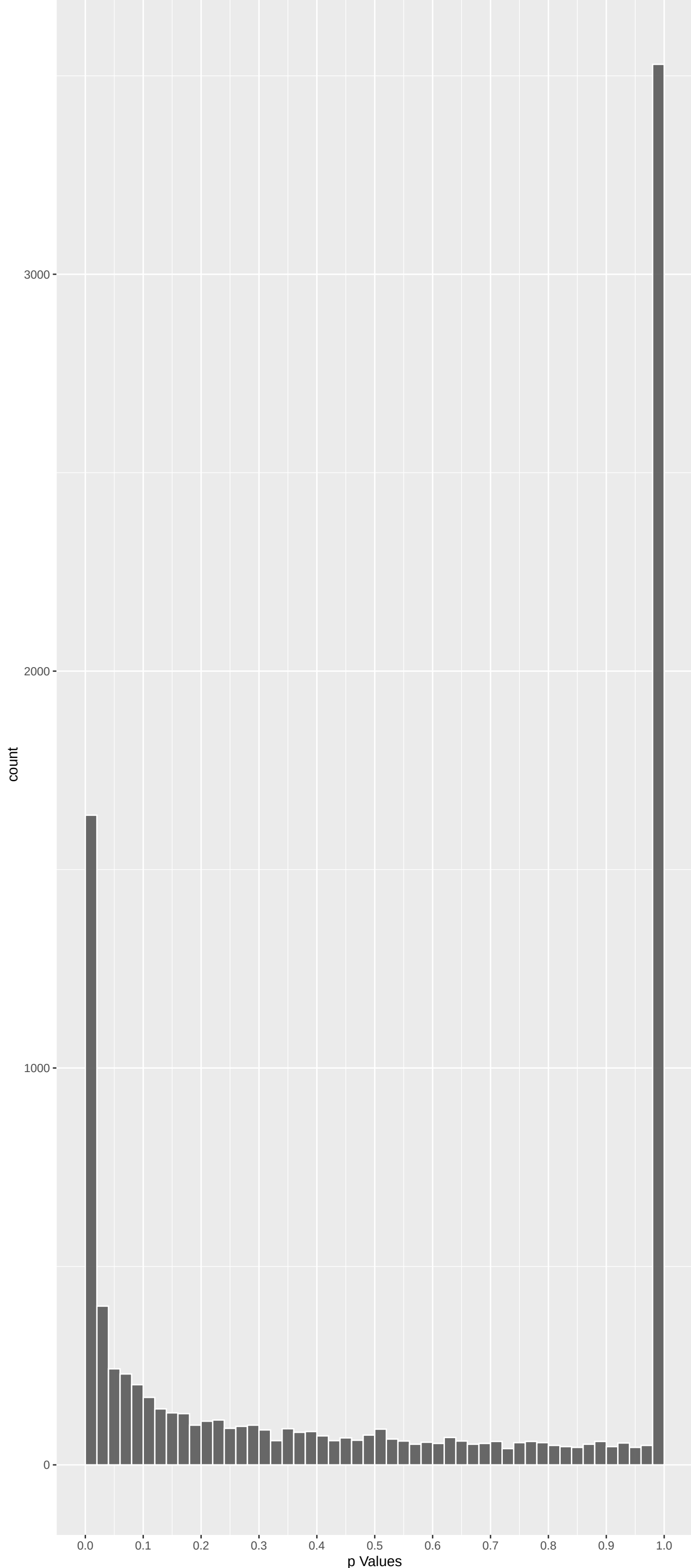
Top Positive genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
PLEC	7.335147	6.644363e-13	3.573e-09	1.921e-05
MMEL1	6.980375	8.831825e-12	1.836e-08	3.291e-05
EP400	6.964074	9.917134e-12	1.836e-08	3.291e-05
TRPM5	7.016268	6.836189e-12	1.836e-08	3.291e-05
TACR2	6.695199	6.461332e-11	8.018e-08	9.950e-05
SFSWAP	6.569396	1.515589e-10	1.358e-07	1.217e-04
ADGRG7	6.238683	1.323811e-09	7.910e-07	4.564e-04
SLC17A4	6.225258	1.442292e-09	8.205e-07	4.564e-04
ITGB4	6.130117	2.634434e-09	1.328e-06	6.657e-04
COL6A1	5.908878	1.033339e-08	3.969e-06	1.524e-03

Top genes by Q-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
SPG11	-7.877562	1.001490e-14	1.616e-10	2.606e-06
PLEC	7.335147	6.644363e-13	3.573e-09	1.921e-05
TMEM131L	-7.338137	6.497642e-13	3.573e-09	1.921e-05
ADGRF1	-7.170844	2.236109e-12	9.018e-09	3.291e-05
CLCNKB	-6.959520	1.024299e-11	1.836e-08	3.291e-05
MMEL1	6.980375	8.831825e-12	1.836e-08	3.291e-05
EP400	6.964074	9.917134e-12	1.836e-08	3.291e-05
TRPM5	7.016268	6.836189e-12	1.836e-08	3.291e-05
SLC14A2	-7.103732	3.642979e-12	1.175e-08	3.291e-05
AGGF1	-6.891699	1.653897e-11	2.668e-08	4.304e-05
PJA2	-6.856022	2.124129e-11	3.115e-08	4.568e-05
ECI2	-6.761969	4.083872e-11	5.490e-08	7.381e-05
TACR2	6.695199	6.461332e-11	8.018e-08	9.950e-05
CLCNKA	-6.668773	7.738535e-11	8.917e-08	1.027e-04
ADGRV1	-6.625078	1.041198e-10	1.120e-07	1.204e-04
ZFAND4	-6.571942	1.489891e-10	1.358e-07	1.217e-04
CFAP20DC	-6.585897	1.356439e-10	1.358e-07	1.217e-04
SFSWAP	6.569396	1.515589e-10	1.358e-07	1.217e-04
TIGD4	-6.535614	1.900473e-10	1.533e-07	1.236e-04
MTBP	-6.544606	1.789571e-10	1.519e-07	1.236e-04
ZNF292	-6.481471	2.724982e-10	2.080e-07	1.525e-04
VEPH1	-6.475435	2.836180e-10	2.080e-07	1.525e-04
RAVER2	-6.310294	8.355200e-10	5.860e-07	4.110e-04
AKAP9	-6.278581	1.025034e-09	6.614e-07	4.134e-04
PPP4R1	-6.271334	1.073902e-09	6.663e-07	4.134e-04

Negative Rho Non-permuted



Top Negative genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
SPG11	-7.877562	1.001490e-14	1.616e-10	2.606e-06
TMEM131L	-7.338137	6.497642e-13	3.573e-09	1.921e-05
ADGRF1	-7.170844	2.236109e-12	9.018e-09	3.291e-05
SLC14A2	-7.103732	3.642979e-12	1.175e-08	3.291e-05
CLCNKB	-6.959520	1.024299e-11	1.836e-08	3.291e-05
AGGF1	-6.891699	1.653897e-11	2.668e-08	4.304e-05
PJA2	-6.856022	2.124129e-11	3.115e-08	4.568e-05
ECI2	-6.761969	4.083872e-11	5.490e-08	7.381e-05
CLCNKA	-6.668773	7.738535e-11	8.917e-08	1.027e-04
ADGRV1	-6.625078	1.041198e-10	1.120e-07	1.204e-04

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Fatty Acid Catabolic Process (GO:0009062)	-0.21385593	58	1.816e-08	9.784e-05	ECI2:8 EHHADH:236 ACAA:300 ACAD11:349 SLC27A4:350 ACOT8:365
Muscle Contraction (GO:0006836)	0.16689190	88	3.770e-08	9.804e-05	TACR2:5 MYH13:31 MYH2:142 MYH8:191 KCNJ12:230 MYH1:421
rRNA Methylation (GO:0030488)	-0.26190903	36	5.459e-08	9.804e-05	TRMT10A:151 TRMT10B:164 FTSJ1:311 THADA:358 TRMT1L:595 TRMO:696
Chemical Synaptic Transmission (GO:00072)	0.09806908	255	7.894e-08	1.063e-04	RIMBP2:27 NPTX1:69 SLC18A2:89 GABRR1:104 RPS6K2:122 NTSR1:133
Regulation Of Heart Rate by Cardiac Cond	0.24580968	39	1.101e-07	1.186e-04	DSG2:212 ISL1:366 CAV1:565 KCNQ1:585 BIN1:694 KCNE1:790
Cardiac Conduction (GO:0061337)	0.21969576	44	4.675e-07	4.196e-04	DSG2:212 ISL1:366 CAV1:565 KCNQ1:585 BIN1:694 KCNE1:790
Fatty Acid Beta-Oxidation (GO:0006635)	-0.20937856	47	6.606e-07	4.449e-04	ECI2:8 AC0X2:88 EHHADH:236 GCDH:273 ACAA:300 ACAD11:349
tRNA Modification (GO:0006400)	-0.17503272	66	5.825e-07	4.449e-04	TRMT10A:151 FTSJ1:311 PUS10:312 THADA:358 DTWD1:439 TRMO:696
Visual Perception (GO:0007601)	0.14552598	95	9.836e-07	5.888e-04	VAX2:38 GNAT2:92 RGS9BP:114 NXNL2:138 CRYAA:145 TRPM1:163
Steroid Metabolic Process (GO:0008202)	-0.17302671	61	3.015e-06	1.624e-03	CYP17A1:28 CYP11A1:60 CYP26C1:99 CYP1A1:124 CYP2R1:146 DHRS4:234
RNA Methylation (GO:0001510)	-0.17508344	58	4.062e-06	1.990e-03	TRMT10A:151 METTL4:206 FTSJ1:311 THADA:358 TFB2M:661 TRMO:696
Striated Muscle Contraction (GO:0006941)	0.18222257	52	5.563e-06	2.498e-03	MYH7B:19 MYH8:191 TNNI3:281 TNNI2:539 KCNQ1:585 GRK2:619
Anterograde Trans-Synaptic Signaling (GO:009678607)	0.09678607	181	7.535e-06	3.123e-03	NPTX1:69 SLC18A2:89 GABRR1:104 RPS6K2:122 NTSR1:133 MTNR1B:153
Modulation Of Chemical Synaptic Transmis	0.11783607	118	1.019e-05	3.920e-03	NPTX1:69 TPRG1L:156 CLSTN1:245 GRM6:440 SLC8A3:531 CPLX3:588
Alpha-Amino Acid Catabolic Process (GO:1	-0.23487968	29	1.207e-05	4.334e-03	TAT:122 SARDH:123 KMO:293 ALDH4A1:355 MCCC1:456 HMGCL1:600
RNA Modification (GO:0009451)	-0.18036546	47	1.909e-05	6.343e-03	FTSJ1:311 DTWD1:439 LCMT2:702 DTWD2:776 PUS3:795 PUS7L:906
Mitochondrial Gene Expression (GO:014005)	-0.12358000	100	2.113e-05	6.343e-03	MRPS10:399 LARS2:485 TFB2M:661 MRPL16:692 PTCO3:750 TFAM:796
Sensory Perception Of Light Stimulus (GO:1	0.12540112	96	2.237e-05	6.343e-03	VAX2:38 GNAT2:92 NXNL2:138 CRYAA:145 TRPM1:163 ARR3:196
rRNA Processing (GO:0008033)	-0.17567427	49	2.124e-05	6.343e-03	FTSJ1:311 DTWD1:439 LCMT2:702 DTWD2:776 PUS3:795 METTL2B:1118
Monocarboxylic Acid Transport (GO:001571)	-0.15305869	63	2.696e-05	7.264e-03	SLC5A12:96 SLC16A5:106 SLC27A4:350 AR14A:369 ATP8B1:449 SLC10A6:589
Mitochondrial Translation (GO:0032543)	-0.12284819	96	3.274e-05	8.400e-03	MRPS10:399 LARS2:485 TSMF:673 MRPL16:692 PTCO3:750 MTIF2:751
Double-Strand Break Repair Via Homologou	-0.11558454	106	4.047e-05	9.911e-03	FANCM:37 SLF1:216 XRCC1:241 MCM9:465 HELQ:537 CDC45:708
Non-Motile Cilium Assembly (GO:1905515)	-0.21049228	31	5.021e-05	1.127e-02	TSHZ3:123 TPRG1L:156 CANCG4:402 GRM6:440 DGKI:800 DRD2:847
Calcium Ion Influx Across Plasma Membran	0.20312820	33	5.420e-05	1.168e-02	ARL13B:170 IFT122:308 SC2D3:371 TGNAR1:677 CEP98:754 ENK1D:912
Regulation Of Chromosome Separation (GO:1	-0.36542097	10	6.301e-05	1.306e-02	SMC4:156 NCAPD2:342 SMC2:354 NUMA1:712 NCAPH:1664 NCAPG:1795
Cardiac Muscle Cell Action Potential Inv	0.21557453	28	7.913e-05	1.550e-02	SCN1A:576 KCNQ1:585 FGFR12:701 KCNJ1:790 RYR2:1131 SCN4B:1221
Cellular Response To Oxygen-Containing C	0.06029899	367	8.056e-05	1.550e-02	ADCV1:25 ILK3:75 RPTOR:152 SLC25A24:167 TGM2:168 ILK6G:204
Adenylyate Cyclase-Modulating G Protein-C	0.09277648	145	1.193e-04	2.143e-02	ADGRG7:7 ADGBR1:21 ADCY1:25 GNAT2:92 VIPR1:136 FFAR3:286
Cellular Response To Acetylcholine (GO:1	0.35151708	10	1.186e-04	2.143e-02	CHRM3:250 SLURP1:259 TYPD1:614 CHRNA2:626 CHRNA2:662 LY6G6D:173
Neuron Projection Morphogenesis (GO:0048	0.09541954	134	1.414e-04	2.457e-02	ADGBR1:21 DBNL:119 CNTN4:165 DHPN2:324 DVNL1:252 IQGAP4:2574
Amino Acid Catabolic Process (GO:0009063)	-0.17641132	38	1.690e-04	2.738e-02	SARDH:123 KMO:293 ALDH4A1:355 MCCC1:456 AMOHD1:573 HIBC8:657
Fatty Acid Oxidation (GO:0019395)	-0.15903802	47	1.634e-04	2.738e-02	ECI2:8 EHHADH:236 ACAA:300 ACAD11:349 HAC1:585 ABCCL1:841
Sister Chromatid Segregation (GO:0000819	-0.18899065	33	1.728e-04	2.738e-02	CHAMP1:78 SMCA:156 KIF18B:503 ESPL1:562 KIF18A:648 KIFC1:818
Cell Junction Assembly (GO:0034329)	0.11829310	83	1.986e-04	3.057e-02	SHANK2:32 CDH26:59 ITGA6:117 DBNL:119 NRCAM:292 SDK1:383
Adenylyate Cyclase-Inhibiting G Protein-C	0.15763554	46	2.184e-04	3.222e-02	CHRM3:250 FFAR3:286 HRRH:318 OPRK1:435 OPRK1:837 DRD2:847
Potassium Ion Transmembrane Transport (G	0.09329398	132	2.213e-04	3.222e-02	KCNV2:20 KCNN3:97 KCNA10:215 KCNJ12:230 KCNQ2:322 KCNN1:372
DNA Repair (GO:0006281)	-0.06455481	274	2.512e-04	3.223e-02	FANCM:37 PARP4:67 MBD4:79 RNF168:87 ERCC6L2:116 RHNO1:138
NADH Dehydrogenase Complex Assembly (GO:	-0.15327488	48	2.412e-04	3.223e-02	EC5IT:98 NDUFS3:376 NDUFAF2:421 NDUFAF1:496 TMEM126A:948 NDUFB8:1145
Mitochondrial Respiratory Chain Complex	-0.15327488	48	2.412e-04	3.223e-02	EC5IT:98 NDUFS3:376 NDUFAF2:421 NDUFAF1:496 TMEM126A:948 NDUFB8:1145

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HAMAI_APOPTOSIS_VIA_TRAIL_UP	-0.10181764	594	3.391e-17	2.200e-13	SPG11:1 AGGF1:6 PJA2:7 CEP192:41 MBD4:79 NARS1:82
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	0.18341469	128	8.135e-13	2.639e-09	NPBWR2:23 CDH26:59 CABLES2:83 DIDO1:102 TCF15:129 NTSR1:133
MARTENS_TRETNINOIN_RESPONSE_UP	0.07997971	676	1.828e-12	3.954e-09	MMEL1:3 COL6A1:11 SHANK2:32 TRIM29:43 PRDM16:63 NPTX1:69
REACTOME_NEURONAL_SYSTEM	0.10230679	383	7.350e-12	1.071e-08	CAMK2B:13 KCNV2:20 ADCY1:25 SHANK2:32 SLC18A2:89 KCNN3:97
REACTOME_MUSCLE_CONTRACTION	0.14513608	187	8.254e-12	1.071e-08	CAMK2B:13 ITPR1:139 MYH8:191 MYBP2C:218 KCNJ12:230 NEB:240
REACTOME_SIGNALING_BY_GPCR	0.07836216	629	2.550e-11	2.757e-08	TACR2:5 TIAM2:12 CAMK2B:13 NPBWR2:23 ADCY1:25 OPN3:62
REACTOME_G_ALPHA_I_SIGNALLING_EVENTS	0.11393493	268	1.496e-10	1.386e-07	CAMK2B:13 NPBWR2:23 ADCY1:25 OPN3:62 GNAT2:92 C3:125
FISCHER_DREAM_TARGETS	-0.06120390	882	9.505e-10	7.709e-07	MTBP:13 FANCM:37 CEP192:41 MDM1:54 DEPDC1:101 WDR76:134
DUTERTRE ESTRADIOL_RESPONSE_24HR_UP	-0.10073451	308	1.330e-09	9.587e-07	KCNK15:21 TET2:93 WDR76:134 CLSPN:136 ZWILCH:196 STL:201
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	0.25499659	46	2.203e-09	1.429e-06	PIEZO1:26 SPTA2L:208 CPNE7:256 ZC3H18:324 TCF25:388 CHMP1A:682
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.07425225	555	2.653e-09	1.565e-06	CAMK2B:13 ADGBR1:21 SLC18A2:89 AATK:113 NTSR1:133 CNTN4:165
REACTOME_DEVELOPMENTAL_BIOLOGY	0.05769274	937	2.981e-09	1.612e-06	COL6A1:11 COL6A2:40 KRT80:64 KRT10:99 RPS6K2:122 DSG3:130
REACTOME_INFECTIOUS_DISEASE	0.06069607	809	5.728e-09	2.859e-06	ADCV1:25 SH3GL3:50 ACE2:51 NOD1:106 C3:125 ITPR1:139
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.27060706	38	7.860e-09	3.469e-06	LFNG:222 MICALL2:496 INTS1:870 CYP2W1:951 FAM20C:1024 ZFAND2A:1085
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	0.09250206	331	8.019e-09	3.469e-06	BNCL1:5 ADGBR1:21 PRDM16:63 VIPR1:136 GATA5:285 TMEM151A:302
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	-0.06819354	596	1.543e-08	6.256e-06	AGGF1:6 LEPROTL1:29 CCG1:36 SCD5:56 APC:107 WDR76:134
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	0.1628636	15	2.375e-08	9.062e-06	EP400:4 SFSWAP:6 GOLGA3:291 PUS1:415 P2RX2:487 MMP17:524
DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN	-0.07589292	449	4.006e-08	1.444e-05	PJA2:7 ZNF292:15 AKAP9:19 MAN2A1:27 DST:35 PAR3:38
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.08762926	331	4.650e-08	1.588e-05	BNCL1:5 ADGBR1:21 PRDM16:63 VIPR1:136 TMEM151A:302 CADM3:305
DACOSTA_UV_RESPONSE_VIA_ERCC3_CN	-0.05592323	824	6.187e-08	2.007e-05	SPG11:1 TMEM131L:2 PJA2:7 ZNF292:15 AKAP9:19 MAN2A1:27
WP_GPCRS_CLASS_A_RHODOPSINLIKE	0.10667564	211	9.737e-08	2.872e-05	NPBWR2:23 OPN3:62 NTSR1:133 MTNR1B:153 OPN1LW:175 BDKRB1:214
NIKOLSKY_BREAST_CANCER_21Q22_AMPLICON	0.41179702	14	9.557e-08	2.872e-05	COL6A1:11 COL6A2:40 SPATCL1:56 DIP2A:242 ADARB1:330 LSS:432
REACTOME_INNATE_IMMUNE_SYSTEM	0.05352342	865	1.151e-07	3.120e-05	NOD1:106 ANPEP:115 DNBAT1:19 RPS6K2:122 C3:125 ITPR1:139
KEGG_TIGHT_JUNCTION	0.13741202	125	1.154e-07	3.120e-05	MYH7B:19 MYH13:31 MYH2:142 AFDN:173 MYH8:191 CLDN22:221
REACTOME_GPCR_LIGAND_BINDING	0.07728889	398	1.340e-07	3.479e-05	TACR2:5 NPBWR2:23 OPN3:62 C3:125 UTSR:132 NTSR1:133
DAZARD_RESPONSE_TO_UV_NHEK_DN	-0.09082150	282	1.645e-07	4.104e-05	TMEM131L:2 AKAP9:19 DST:35 AFFR1:80 TUT4:104 PPRC1:131
BLANCO_MELO_BRONCHIAL_EPITHELIAL_CELLS_I	-0.11873872	163	1.751e-07	4.207e-05	WDR76:134 CLSPN:136 RHNO1:138 ORC1:172 IRAG2:196 EPN3:227
WP_STRATIATED_MUSCLE_CONTRACTION_PATHWAY	0.25717699	34	1.995e-07	4.622e-05	MYH8:191 MYBP2C:218 NEB:240 TNNI7:281 TNNI7:539 DES:569
DODD_NASOPHARYNGEAL_CARCINOMA_DN	-0.04432260	1234	2.246e-07	4.857e-05	FANCM:37 GSTO1:39 CEP192:41 EMILIN2:72 DEPDC1:101 NSD3:108
MARSON_BOUND_BY_E2F4_UNSTIMULATED	-0.05972583	655	2.176e-07	4.857e-05	MTBP:13 FANCM:37 KIAA0825:40 MDM1:54 TBCD1:58 UBAP2:65
BENPORATH_ES_WITH_H3K27ME3	0.04857501	982	3.294e-07	6.891e-05	PLEC:1 CAMK2B:13 BNCL1:5 NKPD1:28 VAX2:38 IGFBP1:46
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	0.08627174	296	3.525e-07	6.891e-05	TACR2:5 NPBWR2:23 OPN3:62 C3:125 UTSR:132 NTSR1:133
REACTOME_CARDIAC_CONDUCTION	0.13446325	120	3.717e-07	6.891e-05	CAMK2B:13 ITPR1:139 KCNJ12:230 ATP1A4:389 CANCG4:402 ATP2A3:510
NUYTEN_EZH2_TARGETS_DN	-0.04942909	938	3.668e-07	6.891e-05	PJA2:7 MTBP:13 LEPROTL1:29 WNK4:47 ABHD11:68 DEPDC1:101
JOHNSTONE_PARVB_TARGETS_3_DN	-0.05385792	784	3.541e-07	6.891e-05	MTBP:13 MAN2A1:27 CCGP1:36 MDM1:54 UBAP2:65 PHF20:91
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	0.10931975	261	3.931e-07	7.084e-05	TACR2:5 NPBWR2:23 GABRR1:104 UTS2R:132 NTSR1:133 VIPR1:136
REACTOME_NEURONTRANSMITTER_RECEPTORS_AND	0.10762113	183	5.338e-07	9.361e-05	CAMK2B:13 ADCY1:25 GABRR1:104 RPS6K2:122 KCNJ12:230 GABRR3:311
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	0.07117631	420	6.196e-07	1.033e-04	CAMK2B:13 FAM83G:16 ADGBR1:21 SLC18A2:89 LC1AM:131 NTSR1:133
RODRIGUES_THYROID_CARCINOMA_ANAPLASTIC_U	-0.05833447	634	6.211e-07	1.033e-04	WDR36:69 DEPDC1:101 APC:107 SMC4:156 ZWILCH:196 STL:201
REACTOME_KERATINIZATION	0.15578602	85	6.984e-07	1.133e-04	KRT80:64 TGM1:99 DSG3:130 KRT13:157 TGM5:166 KRT19:206

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HIV Infections	0.05654629	634	1.516e-06	7.438e-03	PLEC:1 IL16:57 ISG20:58 LPIN1:84 VIPR1:136 IL18:141
Tetany	-0.30936826	21	9.254e-07	7.438e-03	CLDN16:25 TRPM6:45 JMJD1C:190 SLC12A3:472 COMT:646 SLC12A1:705
Glycinuria	-0.35951644	13	7.192e-06	2.352e-02	SLC6A20:209 ALDH4A1:355 PCCA:842 PRODH:893 ACADM:1372 SLC36A2:1597
CATARACT_AUTOSOMAL_DOMINANT	0.39546196	10	1.490e-05	2.436e-02	CRYAA2:76 CRYAA:145 GJA8:228 CRYBB2:380 BFPSP:560 MIP:656
Congenital cerebral hernia	-0.18581002	46	1.315e-05	2.436e-02	CRB1:31 WDPCP:323 RDH12:378 TMEM67:515 ARHGAP31:568 CEP290:606
Mouth Neoplasms	0.12317614	107	1.107e-05	2.436e-02	UPP1:45 IL18:141 PTK2:144 HMGCS2:158 CDKN2B:335 CAV1:565
HYPERGLYCINURIA (disorder)	-0.31797264	15	2.013e-05	2.542e-02	SLC6A20:209 ALDH4A1:355 PCCA:842 PRODH:893 ACADM:1372 SLC36A2:1597
Nuclear cataract	0.21751187	32	2.073e-05	2.542e-02	CRYAA:145 TGM2:168 HSF4:179 GJA8:228 CRYBB2:380 MIP:656
22q11 partial monosomy syndrome	-0.36750784	11	2.439e-05	2.658e-02	JMJD1C:190 COMT:646 HIRA:1086 TBX1:1435 PI4KA:1707 GP1BB:2115
Abnormality of the tonsils	-0.35741233	11	4.053e-05	3.058e-02	JMJD1C:190 COMT:646 HIRA:1086 TBX1:1435 GP1BB:2115 IDUA:2808
Epidermolysis Bullosa Simplex	0.26066410	21	3.559e-05	3.058e-02	PLEC:1 ITGB4:9 KRT80:64 TGM5:166 DES:569 KRT14:597
Hypocalcemia	-0.17798902	45	3.648e-05	3.058e-02	CLDN16:25 TRPM6:45 SLC4A1:139 JMJD1C:190 IFT122:308 COMT:646
Primary microcephaly	-0.11618317	105	4.007e-05	3.058e-02	FANCM:37 TRMT10A:151 WDFY3:154 ORC1:172 STL:201 ERCC2:237
Night blindness, congenital stationary	0.22738136	25	8.341e-05	5.455e-02	TRPM1:163 GPR34:244 PDE6B:339 GRM6:440 FGFR2:542 NYX:549
Complete congenital stationary night bl	0.30411970	14	8.164e-05	5.455e-02	TRPM1:163 PDE6B:339 GRM6:440 FGFR2:542 NYX:549 GPR179:683
Anemia, hereditary spherocytic hemolytic	-0.49058568	5	1.452e-04	6.049e-02	ANK1:32 EPB42:77 SLC4A1:139 SPTA1:140 SPTB:372 NA
Cataract, Central Scaphic, With Sutaral	0.44610166	6	1.542e-04	6.049e-02	GJA8:228 CRYBB2:380 BFPSP:560 MIP:656 CRYBA1:858 CRYGS:2341
Asthma	0.03580659	1060	1.095e-04	6.049e-02	MMEL1:3 TACR2:5 SFSWAP:6 ITGB4:9 ILK3:75 ILK6:57
Dyssebhorric dermatitis	-0.27613601	16	1.314e-04	6.049e-02	JMJD1C:190 COMT:646 HIRA:1086 TBX1:1435 BTD:1826 GP1BB:2115
Familial Intrahepatic cholestasis of pre	-0.44760351	6	1.465e-04	6.049e-02	NR1H4:417 ATP8B1:449 ABCB11:546 VCAAM1:613 ABCB4:1135 TNF:1747
Impaired T cell function	-0.27109497	17	1.091e-04	6.049e-02	JMJD1C:190 COMT:646 UMPS:1069 HIRA:1086 TBX1:1435 CR2:1842
Mildly elevated creatine phosphokinase	0.23078718	23	1.279e-04	6.049e-02	COL6A1:11 COL6A2:40 NEB:240 MYH4:331 CNE:732 LMNA:832
Occipital myelomeningocele	-0.36763360	9	1.340e-04	6.049e-02	J